

Noise Adaptive Quantum Circuit Mapping Using Reinforcement Learning and Graph Neural Network

Vedika Saravanan[✉], *Member, IEEE*, and Samah M. Saeed[✉], *Member, IEEE*

Abstract—To generate physical quantum circuits, quantum gates are often added to the quantum circuit to satisfy the hardware constraints. This process is called quantum circuit mapping. Noise-aware mapping techniques generate physical quantum circuits for noisy intermediate-scale quantum (NISQ) computers. However, the absence of an accurate noise model of the quantum hardware limits its performance. In this article, we propose a noise adaptive quantum circuit mapping approach using reinforcement learning (RL) and graph neural network (GNN)-based reliability predictor. Our RL agent learns a quantum circuit mapping policy that significantly improves the quantum circuit output fidelity by interacting with the environment, which adopts a GNN reliability model that acts as quantum hardware and estimates the physical quantum circuit fidelity. Furthermore, we propose a multi-GNN reliability model to speed up the inference while maintaining high accuracy. Our proposed RL framework fills the gap between the simplified reliability models of the quantum hardware and the realistic noise impact of the quantum hardware on the quantum circuits. We demonstrate the improvement in the output fidelity of quantum circuits generated using our approach compared to other quantum circuit mapping techniques across different real-world quantum hardware using multiple-seed experiments.

Index Terms—Graph neural network (GNN), noise-adaptive quantum circuit mapping, noisy intermediate-scale quantum (NISQ) computer, quantum circuit, reinforcement learning (RL).

I. INTRODUCTION

CURRENT technologies used for building near-term quantum computers referred to as noisy intermediate-scale quantum (NISQ) computers are still in a nascent stage given the limited number of quantum bits (qubits) that are prone to different sources of noise. Significant research and industrial efforts are focused on developing quantum circuit compilation techniques to mitigate errors. Quantum circuit mapping is an essential compilation step to transform logical quantum circuits into physical ones. It targets either minimizing the gate count and the circuit depth or maximizing the fidelity of the physical quantum circuit using a simple noise model

(e.g., [1], [2], [3], [4], [5], and [6]). However, these objectives may provide a very limited improvement in the output state fidelity of the quantum circuit. While recent efforts are focused on modeling quantum hardware noise using deep learning as a promising computationally efficient data-driven approach to predict the output state fidelity of the circuit [7], [8], [9], [10], integrating these techniques with the quantum circuit compilation stack is still unclear. Existing noise-aware quantum circuit mapping techniques do not guarantee significant improvement in the output state fidelity. There is a demand for mapping quantum circuits to NISQ for a broad range of applications that is: 1) compatible with NISQ hardware and constraints; 2) adaptive to frequent changes in physical constraints; and 3) capable of minimizing errors in the quantum circuit to achieve high output state fidelity.

To address these challenges, we propose a noise adaptive quantum circuit mapping approach based on principles of reinforcement learning (RL) and graph neural networks (GNN). To the best of our knowledge this is the first work that shows how to integrate the GNN reliability model into the quantum circuit compilation stack to boost the output state fidelity in a cost-effective way. Our proposed framework explores quantum circuit design space to maximize the output state fidelity of the quantum circuit given the physical constraints of the quantum hardware. Our proposed framework provides an accurate performance prediction of the resulting physical quantum circuit using a GNN-based fidelity predictor. The key features of our RL framework are as follows: 1) it runs inference during the RL training phase on the trained GNN model to enable real-time assessment of the fidelity of the resulting physical quantum circuit without any interaction (execution) with the quantum hardware and 2) it is trained once but can be applied: a) across different calibration windows/cycles and b) on unseen quantum circuit designs. The contributions of our work are as follows.

- 1) We propose a noise-adaptive quantum circuit mapping based on RL and GNN models to boost the output state fidelity of the circuit. By incorporating a GNN-based fidelity predictor into our RL framework, we accurately model the impact of the quantum hardware noise on the circuit, and therefore, generate physical quantum circuits that mitigate errors.
- 2) We propose a multilevel GNN-based fidelity predictor to improve the scalability of our GNN models while maintaining high prediction accuracy.

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The authors are with the Department of Electrical Engineering, The City College of New York, New York, NY 10031 USA (e-mail: ssaeed@ccny.cuny.edu).

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- 3) We propose to use node embedding to encode not only the structure of the current state of the RL agent but also the backend features (e.g., error rates) given a selected initial qubit allocation.
- 4) We conduct an extensive evaluation of the proposed framework using different quantum benchmark circuits executed on different real-world quantum computers with 7 and 16 qubits.
- 5) We demonstrate the efficiency of our proposed framework compared to the state-of-the-art noise-aware quantum circuit mapping (i.e., Qiskit [1], t|ket [6], and noise adaptive mapping [4]) and recently proposed RL-based quantum circuit mapping approaches [11], [12].

The remainder of this article is organized as follows. Section II provides an introduction to quantum circuit mapping, quantum hardware errors, GNN models, and RL techniques. Section III presents related works on noise-aware quantum circuit mapping, machine learning (ML)-assisted quantum circuit mapping, and ML for modeling the reliability of quantum circuits. We motivate the use of ML-assisted noise modeling in quantum circuit mapping in Section IV. We describe our proposed RL framework for quantum circuit mapping in Section V. We evaluate our proposed RL framework using several benchmark circuits in Section VI. We discuss the efficiency of our proposed framework in Section VII. We conclude this article in Section VIII.

II. BACKGROUND

A. Quantum Circuit Mapping

Quantum algorithms are implemented using quantum circuits that are executed on quantum hardware. A quantum circuit applies a series of gates, which range from single-qubit to arbitrary gates, to the qubits. An arbitrary n -qubit gate is decomposed into single- and two-qubit gates. Many of the current quantum hardware restrict the qubits' interaction with each other (e.g., superconducting quantum hardware), which is described as the coupling graph (CG), in which each node is a qubit and each edge describes a permissible two-qubit gate. To enable any two-qubit gates in the circuit, a *SWAP* gate is typically used which consists of three controlled not (CNOT) two-qubit gates that switch the state of the two qubits. Quantum circuit mapping aims at minimizing the number of required *SWAP* gates to build a physical quantum circuit that is compatible with the constraints of the quantum hardware (e.g., [2], [3], and [5]).

Example 1: An example of quantum circuit mapping is shown in Fig. 1, which adds five *SWAP* gates to satisfy the coupling constraint of the IBM Q Jakarta quantum computer.

B. Errors in Quantum Hardware

Different error types are encountered during the quantum circuit execution on NISQ hardware. Qubits suffer from unexpected and uncontrolled interaction with the environment. Operations applied to the qubits cause active errors, namely, one- and two-qubit gate errors and readout errors. Idle qubits are also prone to other sources of noise that emerge from uncontrollable environmental fluctuations resulting in idling

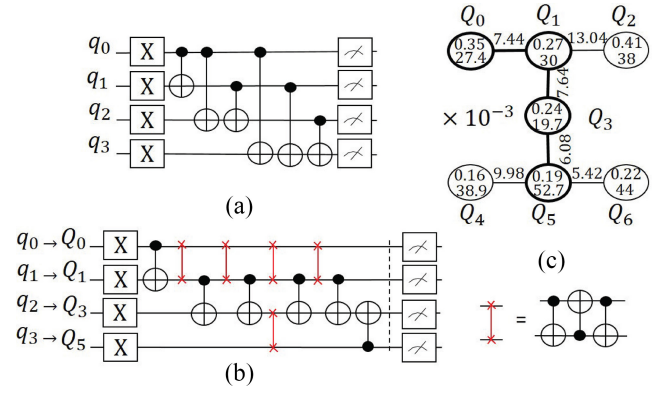


Fig. 1. Logical quantum circuit (a) maps to a physical one (b) which satisfies the coupling constraint (c) of IBM Q Jakarta. *SWAP* gates are marked in the red color.

errors. Undesired interactions between coupled qubits lead to crosstalk errors. Quantum decoherence causes the loss of qubit information as well due to the qubit sensitivity to the environment.

Example 2: Fig. 1(c) shows an example of the error rates of IBM Q Jakarta CG. The upper and the lower numbers of each node in the graph represent the single-qubit gate and readout errors of the corresponding qubit, while the number on each edge of the graph represents the two-qubit gate error of the corresponding CNOT gate.

To assess the output fidelity of the quantum circuit executed on NISQ devices, different fidelity metrics can be used, such as the probability of successful trail (PST) for single-output circuits and the Hellinger fidelity (HF) for multi-output circuits defined as follows:

$$\text{PST} = \frac{\text{Number Of Successful Trials}}{\text{Total Number Of Trials}}$$

$$\text{HF} = \left(1 - \frac{1}{2} \sum_{i=1}^m (\sqrt{s_i} - \sqrt{e_i})^2\right)^2$$

The HF is computed based on the Hellinger distance between the noise-free ($s_1, s_2, \dots, s_m$) and noisy ($e_1, e_2, \dots, e_m$) output distributions of the quantum circuit.

C. GNN

GNNs transform graphs with their node- and edge-related features through embeddings to infer their data. The embedding can be learned for each node, edge, or the entire graph depending on the GNN task (i.e., graph classification, node classification, link prediction, and graph clustering). Node embedding maps the nodes of the graph to the embedding space so that the similarity in the embedding space approximates the similarity in the network. To ensure that the embedding incorporates the graph connectivity, message passing is applied, which exchanges neighboring nodes' features. A popular GNN model is sample and aggregate (GraphSAGE) [13]. It uses an inductive learning technique, and hence, it can predict the node embedding of new unseen graphs. The embedding is conducted in two major steps, aggregate and update. Different types of aggregation functions

can be applied to learn the neighboring nodes (e.g., mean, max-pooling, and GCN [14]).

D. RL

RL is an ML approach that trains an agent to make the right decisions to maximize the reward of a given problem by exchanging information with the environment. It consists of a set of states, actions, transition probabilities, and rewards known as a Markov decision process. In each time step, an agent takes an action, which the environment evaluates given the current state and produces a reward accordingly. The agent continues to take action until the goal is reached. A sequence of states and actions is referred to as an episode. Through many interactions with the environment in different episodes, the agent learns how to maximize the reward.

Several RL algorithms have been proposed to solve the decision-making problem, such as value-based (e.g., [15]) and policy-based RL models (e.g., [16]). In the value-based RL models, given a state (s) and an action (a), the agent tries to learn the optimal action-value function $Q^*(s, a)$ (Q -value), which drives the policy to maximize the reward. In policy-based RL algorithms, the policy is manipulated directly to obtain the optimal policy for a given state $\pi(a | s)$. Policy-based RL algorithms are more stable and reliable in converging to a good solution. Value- and policy-based RL algorithms are considered model-free RL, which assumes no prior knowledge of the environment.

III. RELATED WORK

A. Noise-Aware Quantum Circuit Mapping

Several noise-aware quantum circuit mapping approaches, which are sensitive to the variation in the error rate of the quantum hardware have been proposed to maximize the output fidelity of quantum circuits (e.g., [4] and [17]). They take into account the physical properties of the quantum hardware (i.e., error rates, gate time, and coherence time) computed in each calibration to allocate the physical qubits and schedule the quantum gates. Accordingly, a simplified reliability model (such as estimated success probability (ESP) [18]) has been used to guide heuristic (e.g., [1], [4], and [6]) and optimal [4], [17] quantum circuit mapping algorithms. Among these techniques, the proposed work in [4] has provided a Satisfiable problem solver-based mapping approach and a greedy scalable solution that prioritizes placing qubit pairs with the most frequent CNOT interactions. Qiskit adopts a stochastic approach for quantum circuit mapping [1]. |ket) uses a noise-aware qubit allocation method and relies on a heuristic algorithm to schedule quantum gates, which minimizes the number of SWAPs [6]. While all these approaches are efficient in terms of execution time and the required SWAP overhead to generate physical quantum circuits, they use simple fidelity/reliability metrics that do not capture the different sources of noise in NISQ hardware.

B. ML-Assisted Quantum Circuit Mapping

ML has been applied in different contexts to enable efficient quantum circuit mapping. ML has been integrated with a

heuristic-based technique for mapping quantum circuits to speed up the search for the initial qubit assignment that minimizes the circuit depth [19]. A deep neural network has been used to find the initial qubit assignment to maximize the output fidelity by training the model with features extracted from the circuit and the quantum computer [20]. RL techniques for quantum circuit mapping have been proposed with different quantum circuit encoding techniques to reduce either the circuit depth [11], the gate count [12], or multiple objectives including a simplified fidelity metric [21]. Although several quantum circuit mapping approaches have been proposed, achieving high output state fidelity is still challenging.

Other approaches target a noise-aware ansatz construction for variational quantum algorithms [22], [23]. The ansatz is generated either using a neural architecture search to find the best performance ansatz based on a noisy quantum circuit simulator [23] or by designing subcircuits that satisfy the coupling constraint of the quantum hardware as building blocks to generate the ansatz, and therefore avoid using SWAP gates in the subcircuits [22]. In our work, we 1) accurately model the noise of the quantum hardware without using any noisy simulators which do not scale to large quantum circuits with arbitrary gates and 2) target general quantum circuits, which can be useful for different classes of quantum algorithms (e.g., Trotterization algorithms that emulate the actual quantum evolution without using a hybrid classical-quantum method [24]). Our work can also be integrated with different quantum circuit optimization techniques.

C. ML for Quantum Circuit Reliability Modeling

ML has been utilized to model the reliability of quantum circuits [7], [8], [9], [25]. Some ML models consider the quantum hardware as a black box and thus rely on frequent training to ensure high prediction accuracy [7]. Others use the physical constraints of the quantum hardware and its error rates computed during the calibration process as features [8], [9], [26]. To ensure high accuracy, ML models either require runtime data collected by executing test circuits on the quantum hardware [8] or capture the circuit structure using GNN [9], [10]. The input to the GNN model is the gate dependency graph of the quantum circuit, in which nodes are gates and edges are direct dependencies between gates. GNN is shown to provide superior results compared to other ML-based reliability models [9], [10], [26]. While many ML techniques predict the reliability of quantum circuits at the physical level after conducting the mapping, ML can also predict the reliability of the quantum circuits executed on different quantum hardware at the logical level as a means of projecting the system-level performance of the quantum hardware [25].

IV. MOTIVATION

Existing objectives for quantum circuit mapping may fail to maximize the output fidelity of circuits executed on NISQ computer.

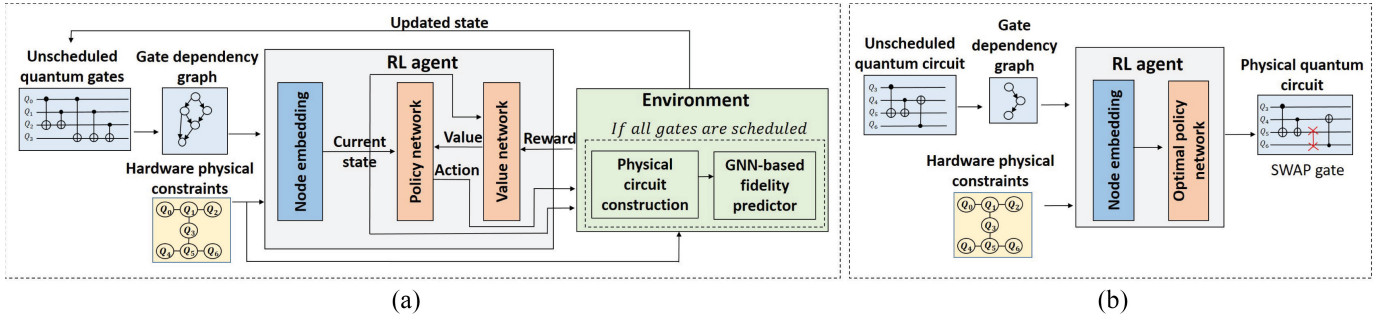


Fig. 2. Overview of the (a) training and (b) inference of our proposed RL-based noise adaptive quantum circuit mapping.

TABLE I
QUANTUM CIRCUIT MAPPINGS EXECUTED ON IBM Q JAKARTA

Circuit	PST	D	#CNOT	ESP
M_1	73.05	28	19	0.72
M_2	67.58	25	27	0.68
M_3	70.90	36	17	0.74
M_4	65.14	35	21	0.72
M_5	67.87	25	23	0.70

Example 3: Let us consider the logical quantum circuit shown in Fig. 1(a). We generate five different physical implementations (mappings) of the circuit and execute them on IBM Q Jakarta during the same calibration window (the period after quantum hardware is calibrated until the next calibration). All the physical quantum circuits share the same initial qubit assignment. Table I shows the output fidelity percentage PST, the circuit depth (D), the number of CNOT gates ($\#CNOT$), and the ESP computed by multiplying the error rates of the circuit gates and the readout operations. While D , $\#CNOT$, and ESP are typical objectives for quantum circuit mapping, optimizing any of these objectives does not maximize the output fidelity. Specifically, physical quantum circuit M_1 has the highest output fidelity but neither its D nor its $\#CNOT$ is the minimum. The ESP of M_1 is not the maximum either.

While the advantage of utilizing GNN models in accurately projecting the quantum circuit output fidelity has been recently shown, these models have not been put to use in the quantum circuit mapping to maximize the output state fidelity of the quantum circuit.

V. PROPOSED FRAMEWORK

A. Overview

We solve gate scheduling for quantum circuit mapping by formulating a policy-based RL problem. The agent is the quantum circuit scheduler and the environment consists of a GNN-based fidelity predictor (Fid_pred) of the physical quantum circuit. The goal of the RL algorithm is to ensure that the agent learns an optimal policy π to generate physical quantum circuits with a very high output fidelity by interacting with the GNN-based fidelity predictor (environment). Fig. 2 shows the overview of our proposed framework. Graph embedding through node embedding is applied to two components of our proposed framework. In the environment, node embedding is used by the GNN-based fidelity predictor to represent

the physical quantum circuit. At the agent, node embedding is conducted to represent the current state of the quantum circuit. The use of node embedding enables encoding different quantum circuits which are represented as graphs of different numbers of nodes and edges.

B. RL for Quantum Circuit Mapping

The main components of our RL are as follows.

- 1) *State:* s_t is the current state that is described as a gate dependency graph of the unscheduled gates of the quantum circuit at time t , which does not satisfy the coupling constraint of the quantum hardware.
- 2) *Action:* a_t is the SWAP gate applied to the circuit at time t . It corresponds to one edge in the quantum hardware CG.
- 3) *State Transition:* P is the probability that the action a_t in state s_t will result in state s_{t+1} . If the selected SWAP gate at time t (a_t) leads to at least one executable layer (L_t) in the quantum circuit, we update s_t to become the remaining unscheduled quantum circuit layers after applying the SWAP gate (a_t) using an update function. Otherwise, the state remains the same

$$s_{t+1} = \begin{cases} \text{Update}(s_t, a_t) - L_t, & \text{if } L_t \neq \phi \ \forall (s_t, a_t) \\ s_t, & \text{otherwise.} \end{cases}$$

- 4) *Reward:* $R(s_t, a_t)$ is the reward of taking action a_t in state s_t . In this problem, we set the reward to 0 for all intermediate actions, except the action at the end of the episode where the reward is the predicted output state fidelity provided by the GNN-based fidelity predictor (Fid_pred) for the resulting physical quantum circuit (C) after applying all the actions of the episode. This is a feasible assumption since the output fidelity of the quantum circuit depends on the entire circuit structure due to the complex noise modeling of the hardware (e.g., crosstalk errors and decoherence errors). We also use $\alpha < 0$ as a penalty for actions (SWAP gates) that lead to no executable layers

$$R(s_t, a_t) = \begin{cases} \text{Fid_pred}(C), & \text{if } s_{t+1} = \phi \\ 0, & \text{if } (s_{t+1} \neq \phi) \wedge L_t \neq \phi \\ \alpha, & \text{if } (s_{t+1} \neq \phi) \wedge L_t = \phi. \end{cases}$$

To generate a fixed low-dimensional vector for any circuit structure that represents the current state, we apply embedding.

Algorithm 1: Environment

Input: s_t, a_t, A, C'
Output: $s_{t+1}, R(s_t, a_t)$

```

1  $s_{temp} = \text{Update}(s_t, a_t);$  // Apply the SWAP
2  $N = \text{Layers\_number}(s_{temp}); L_t = \phi, R(s_t, a_t) = 0$ 
3 for  $i = 0$  to  $N$  do
4   if  $\text{Executable}(l_i)$  then
5      $L_t = L_t \cup \{l_i\};$ 
6     if  $l_i == N$  then // End of the episode
7        $C = \text{Build\_physical\_circuit}(A \cup \{a_t\}, C');$ 
8        $R(s_t, a_t) = \text{Fid\_pred}(C);$ 
9     else
10      break;
11 if  $L_t \neq \phi$  then
12    $s_{t+1} = s_{temp} - L_t$  // Exclude  $L_t$ ;
13 else
14    $s_{t+1} = s_t;$ 
15    $R(s_t, a_t) = \alpha;$ 

```

Unlike existing work, we do not only embed the circuit structure but also its physical properties given the current qubit assignment. To enable inference across different calibration windows, we add as node features, the gate error rate, the gate's qubit id (s), and their coherence time obtained from the back-end properties. To penalize for the nonadjacent two-qubit interaction, we identify the shortest path between the nonadjacent two qubits ($D(i, j)$) using Dijkstra's algorithm to compute the gate error as the summation of the error rates of all the two-qubit gates including *SWAP* gates required to apply the nonadjacent two-qubit gate as $\sum_{g_k \in D(i, j)} e_{g_k}$. As the current state includes physical constraints generated during the calibration process, which may fail to accurately model the errors in the quantum circuit, we learn the effect of these error rates on the output fidelity of the circuit by interacting with the environment (the GNN-based fidelity predictor). The generated fixed feature vector of the current state using graph embedding is concatenated with the circuit's main features, including the circuit depth and the single-qubit and the CNOT gate count.

We use the proximal policy optimization (PPO) [16] RL algorithm to train our agent and update the policy and the value network parameters. The RL agent takes actions in the form of *SWAP* gates until the entire quantum circuit satisfies the coupling constraint of the hardware, which indicates that the episode has ended.

Example 4: Fig. 3 shows an example of applying a *SWAP* (Q_6, Q_7) gate at time t (a_t) to an unscheduled quantum circuit (s_t) provided in Fig. 3(b) to be executed on IBM Q Guadalupe with an initial qubit allocation of $\{Q_4, Q_6, Q_7, Q_{10}, Q_{12}\}$. The resulting updated unscheduled quantum circuit layers (s_{t+1}) after excluding all the executable layers (L_t) is provided in Fig. 3(c).

C. Integration of GNN-Based Fidelity Predictor With RL

We simulate the environment as shown in Algorithm 1, which computes the reward and the next (updated) state. Algorithm 1 takes as an input (s_t, a_t), the list of all valid actions of the episode so far (A), and the original quantum

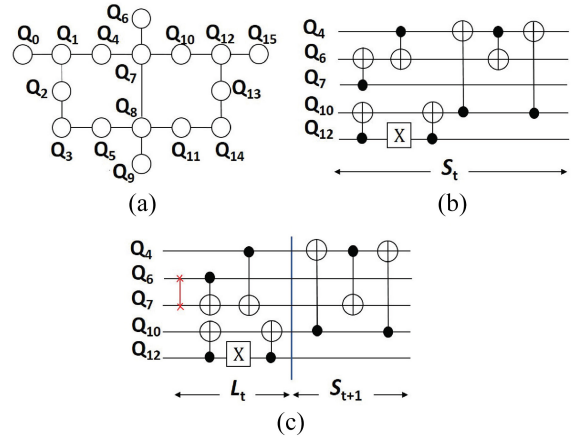


Fig. 3. (a) IBM Q Guadalupe CG, (b) unscheduled quantum circuit, and (c) its corresponding implementation after applying *SWAP*(Q_6, Q_7).

circuit (C'). The algorithm constructs the physical quantum circuit (C) and feeds it to the GNN-based fidelity predictor along with the physical constraints of the quantum hardware at the end of the episode to predict the output fidelity (reward) during the RL agent training phase.

D. GNN-Based Fidelity Predictor

We adopt the GNN-based fidelity predictor since it has proven to provide a better reliability model than other data-driven reliability models due to its ability to capture the circuit structure and its rich feature set at the gate level [9]. The input to the GNN model is in the form of a gate dependency graph of the physical quantum circuit. Each node (gate) has a technology-dependent feature set that enables bridging the gap between the quantum hardware error rates computed during the calibration process and their impact on the circuit output fidelity for accurate noise modeling. Our gate features include the qubit id (s) of the gate (Q_{id}), the gate type (*Type*), the error rate (E_g), the time (T_g), circuit layer (L_g), the parallel gates number in the same gate layer (P_g), and the coherence time ($T1_g, T2_g$) of the qubits of the gates. Embedding is also applied to enable information sharing across neighboring nodes. Given that node embedding enables message passing of neighboring nodes (gates) within a short distance in the gate dependency graph, we extend the gate features of the existing GNN reliability predictor to include the number of all subsequent dependent gates (reachable gates in the dependency graph) that are directly or indirectly connected to the current gate through two-qubit gates (F_{out}). This feature is inspired by our recent work that shows the effectiveness of considering the number of reachable gates in modeling the reliability of quantum circuits [27]. The graph feature vector is provided as an input to a neural network to conduct a regression-based graph classification that predicts the output fidelity of the physical quantum circuit.

By incorporating the physical properties of the gates and their qubits, the GNN-based fidelity predictor learns the impact of the reported error rates of any calibration process on the quantum circuit output fidelity. Thus, our GNN-based fidelity

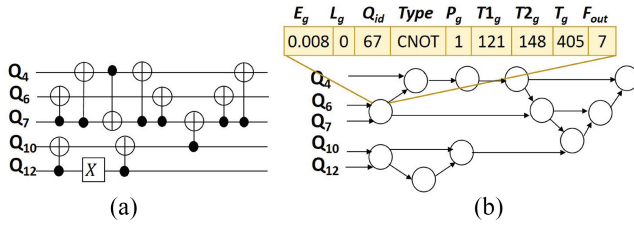


Fig. 4. (a) Physical quantum circuit that satisfies the IBM Q Guadalupe CG and (b) its gate dependency graph.

predictor can be used to infer the output fidelity of quantum circuits to be executed on different new calibration windows.

Example 5: Fig. 4 illustrates a physical quantum circuit implementation that satisfies the CG of IBM Q Guadalupe and its corresponding gate dependency graph. The figure also shows an example of all the node features for a single node in the gate dependency graph.

E. Multilevel GNN-Based Fidelity Predictor

Since the runtime and the accuracy of GNN models are major challenges for large-scale quantum computers, we propose a multilevel GNN-based fidelity predictor as a bottom-up approach to provide an expressive representation of the training dataset and the embedding for large-scale quantum hardware in a cost-effective way. Our proposed multilevel GNN-based fidelity predictor reduces the amount of message passing required for graph embedding while maintaining the circuit topology and a high prediction accuracy, which accelerates the runtime of the GNN-based fidelity predictor. We can further speed up our proposed noise-adaptive quantum circuit mapping by enabling parallel and distributed processing of the fidelity predictor.

Our approach entitles partitioning the qubits of the quantum hardware into regions, using a graph partitioning algorithm (e.g., [28] or [29]). Local GNN (LGNN) models are used each operating on a different subset of qubits of the quantum hardware to learn the node embedding and accordingly the noise model of each region. The number of LGNN models is fixed per quantum architecture/computer. A global GNN (GGNN) is applied next, which combines the learning outcome of different regions in addition to the global circuit structural information to make the final decision.

To infer the output fidelity of a quantum circuit using a multilevel GNN-based fidelity predictor we apply the following method. If all the qubits of the physical quantum circuit belong to a single region, its LGNN will be used. Otherwise, the circuit will be partitioned into subcircuits (blocks) each belonging to one qubit region. The LGNN models will be applied first to the corresponding subcircuits similar to the standard GNN-based fidelity predictor discussed in the previous subsection. The GGNN is used next to model the interaction between different blocks of the quantum circuits. The GGNN model takes as an input the block dependency graph of the entire quantum circuit, which consists of nodes for different subcircuits and elementary gates that connect the blocks. The block dependency graph

Algorithm 2: Quantum Circuit Partitioning

Input: C, N, RQ, Q
Output: PR, V

```

1  $Pr[i] = \phi, \forall i \in N$ 
2  $L[i] = 0, \forall i \in Q$ 
3  $Used[i] = 0 \forall i \in Q$ 
4 for each  $g_i$  in  $C$  do
5   if  $|g_i| == 2$  then
6      $(q_1, q_2) = \text{Qubits}(g_i)$ 
7     if  $RQ(q_1) == RQ(q_2)$  then
8        $k = \max(L(q_1), L(q_2));$ 
9        $r = RQ(q_1);$ 
10      Add  $g_i$  to  $Pr[r][k]$ 
11       $L(q_1) = k; L(q_2) = k;$ 
12       $Used(q_1) = 1; Used(q_2) = 1;$ 
13   else
14     Add  $g_i$  to  $V;$ 
15     if  $Used(q_1)$  then
16        $L(q_1) ++;$ 
17     if  $Used(q_2)$  then
18        $L(q_2) ++;$ 
19      $Used(q_1) = 0; Used(q_2) = 0;$ 
20   else
21      $q_1 = \text{Qubits}(g_i);$ 
22      $r = RQ(q_1);$ 
23     Add  $g_i$  to  $Pr[r][L(q_1)]$ 
24      $Used(q_1) = 1;$ 

```

incorporates not only the structure of the entire quantum circuit but also the noise parameters associated with the entire circuit structure. Each node of the block dependency graph includes most of the typical GNN node features (E_g , L_g , $Type$, P_g , $T1_g$, and $T2_g$ on average, and T_g and F_{out} for the subcircuit), its LGNN fidelity (Fid_{LGNN}), and the block region ID (ID_B). The LGNN fidelity is the predicted fidelity of the subcircuit using the corresponding LGNN models.

Quantum circuit partitioning has been applied to support scalable quantum circuit synthesis (e.g., [30]). Unlike the existing partitioning approach, our main constraint is, instead of the maximum number of qubits in each subcircuit, the qubit IDs of the subcircuit, which should all belong to one of the quantum hardware regions. Algorithm 2 illustrates how the circuit partitioning is conducted. The input to the algorithm is the physical quantum circuit (C), the number of quantum hardware regions (N), the list of qubits in each region (RQ), and the number of hardware qubits (Q). The algorithm outputs the list of subcircuits under each hardware region (Pr); subcircuit i of region j is represented as $Pr[j][i]$. It also generates the list of gates that violate the quantum hardware region boundary (V), i.e., a set of gates that connect qubits from different hardware regions. In the algorithm, we trace the circuit gate by gate, identify the hardware region of the qubit(s) in each gate, and assign the gate to a subcircuit of the corresponding region. When a gate that interacts with more than one hardware region is encountered, the corresponding qubits are excluded from their subcircuits and included in the next subcircuits of these regions. We store the last subcircuit id of each qubit used in each gate in (L) to keep track of the subcircuit boundary (last gate applied to each qubit in

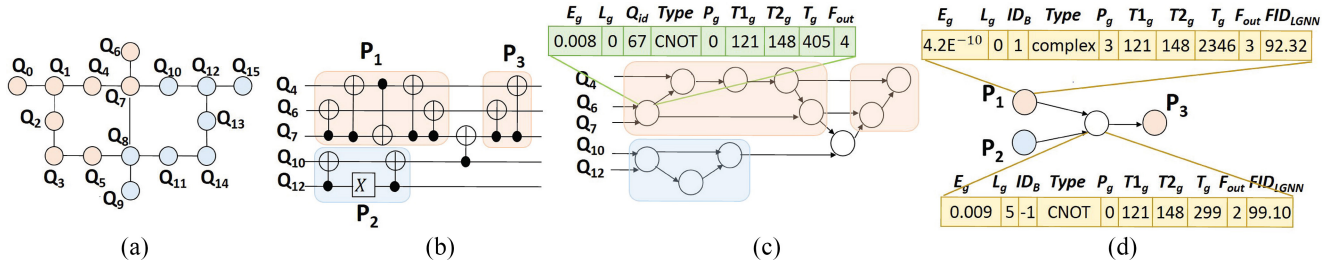


Fig. 5. (a) IBM Q Guadalupe CG regions. (b) Partitioned physical quantum circuit according to (a). (c) Gate dependency graph of each subcircuit. (d) Block dependency graph of the entire circuit.

a given subcircuit). To ensure updating L correctly for each qubit, *Used* is utilized to check whether the last gate applied to the qubit does not violate the quantum hardware region boundary.

Example 6: Fig. 5 shows the coupling constraint of IBM Q Guadalupe quantum hardware that is partitioned into two regions of different colors, the same physical quantum circuit provided in Fig. 4(a) partitioned into different subcircuits (blocks), the gate dependency graph of each subcircuit and its node features, and the block dependency graph of the entire quantum circuit and its node features. The example also shows the value of different node features at the gate and the block dependency graphs. Note that the white node in the block dependency graph in Fig. 5(d) represents a gate that connects different quantum hardware regions in which Fid_{LGNN} is the corresponding gate fidelity based on the gate error.

VI. EXPERIMENTAL EVALUATION

A. Experimental Setup

We evaluate our RL model on real-world quantum computers: 7-qubit IBM Q Jakarta and 16-qubit IBM Q Guadalupe.¹ We use GraphSAGE GNN-based method for predicting the fidelity [31]. We train the GNN model first using a diverse training dataset ranging from 3016 to 4200 random quantum circuits with 4–10 qubits collected over a period of 7 and 8 months for IBM Q Jakarta and IBM Q Guadalupe, respectively. We implemented a script for random quantum circuit generation. The random circuits consist of universal gates circuits (RC_1) based on the concept of un-compute [32], [33] and Clifford gates-only circuits (RC_2). The random circuits were executed through different calibration windows. They produce either single- or multi-outputs. The PST fidelity metric is used for single-output quantum circuits, which GNN-based fidelity predictor is trained using RC_1 , while the HF fidelity metric is used for multi-output quantum circuits, which GNN-based fidelity predictor is trained using a combination of RC_1 and RC_2 .² The GNN models are built using 512 batch sizes and 300 epochs. We use a single GNN (SGNN) with three hidden layers to generate a fixed dimensional vector of size 256 for the gate dependency graph, which is an input

to the conventional neural network with two fully connected layers to predict the output fidelity of quantum circuits. The selected model architecture has shown to provide very high prediction accuracy. We also built a multilevel GNN-based fidelity predictor (MGNN), in which the qubits are partitioned into two regions of equal qubit numbers for IBM Q Guadalupe or almost equal for IBM Q Jakarta. Thus, we build two LGNN models and a global one. The two-level MGNN is selected to enable propagating neighboring nodes' features within more than a two-hop distance in a cost-effective way. We divide our training circuits into three groups, one for each LGNN and one for the GGNN depending on their qubits. We use two hidden layers for the LGNN models and one hidden layer for the GGNN model. We implement our RL model using the PPO algorithm in Tianshou RL library [34] based on PyTorch. We use one hidden layer for both the value and the policy network to guide the agent's actions that maximize its long-term cumulative reward. Graph embedding performed at the RL agent to embed the current state of the quantum circuit into a fixed feature vector uses one hidden layer. We use 1000 datasets of random quantum circuits and 500 episodes to train each RL model. All the models have been trained on Ubuntu 16.04.6 server with Intel Xeon CPU E5-2620 v4 @ 2.10 GHz and four Nvidia GeForce GTX 1080 GPUs. We use four GPUs for training each model.

We use benchmark circuits with 5–15 qubits to test our GNN-based fidelity predictor and RL models. We use `mod5d1_63`, `4g13_92`, `4mod5-v0_19`, `4mod5-v0_20`, `4mod5-v1_22`, `tof_4`, `mod5mils_65`, `mod_mult_55`, and `mod5d1_63` from RevLib [35], quantum Fourier transform (QFT) [36] from Qiskit [1], quantum phase estimation (QPEA), Mermin-Bell (MB), variational quantum eigen solver (VQE), and vanilla-quantum approximate optimization algorithm (V_QAOA) from SupermarQ [37]. We also use QAOA circuits from Cirq [38] to solve Max-Cut problem for different graphs, including QAOA_1 for 15 nodes with at most five edges per node, and QAOA_2 with 13 nodes with at most six edges per node. Table II describes the properties of the benchmark circuits before mapping, which include the number of qubits (# Qubits), the number of single-qubit (SQG) and CNOT gates (CNOT), the number of measurement operations (M), the circuit depth (Depth), and the expected output (Exp. Output). All the selected benchmark circuits do not satisfy the coupling constraint of the hardware to show the effectiveness of our proposed RL. Due to the limited access to the quantum hardware, only a subset of these benchmarks is considered

¹ We demonstrate the effectiveness of our RL using quantum computers of a small number of qubits due to the limited access to larger quantum hardware.

² While we use two different fidelity metrics to show that our proposed framework is applicable to any fidelity metric, HF metric can be used for both single- and multi-output quantum circuits.

TABLE II
PROPERTIES OF QUANTUM CIRCUIT BENCHMARKS

Benc- hmark	# Qubits	#			Depth	Exp. Output
		SGQ	CNOT	M		
RC ₁	4-10	25-350	36-238	4-10	43-285	All 0's
RC ₂	4-10	37-114	13-40	4-10	23-130	NA
mod5d1_63	5	9	13	5	14	00000
4mod5-v0_19	5	19	16	5	22	11000
4mod5-v0_20	5	10	10	5	13	10111
QPEA	5	56	42	4	40	0011
QFT	6	57	39	6	44	00000
mod5mils_65	5	19	16	5	22	11000
4mod5-v1_22	5	10	11	5	13	10000
4gt13_92	5	36	30	5	39	00000
mod_mult_55	9	71	48	9	52	000001000
tof_4	7	45	30	7	52	0000000
MB	7	53	39	7	46	1000000
VQE	5	20	10	4	11	NA
V_QAOA	7	35	42	7	42	NA
QAOA_1	15	302	111	15	153	NA
QAOA_2	13	113	76	13	89	NA

TABLE III
INFERENCE TIME AND ACCURACY (RMSE) OF
SGNN AND MGNN FOR IBM Q JAKARTA

Benchmark	Time (μ s)		RMSE	
	SGNN	MGNN	SGNN	MGNN
4mod5-v0_20	0.98	0.16	1.15E-02	1.62E-02
4mod5-v0_19	0.88	0.05	1.38E-02	1.84E-02
4mod5-v1_22	1.09	0.27	1.36E-02	1.83E-02
4gt13_92	1.06	0.23	1.66E-02	2.12E-02
mod5d1_63	1.25	0.47	1.57E-02	2.04E-02
V_QAOA	1.87	0.11	2.14E-02	2.61E-02
tof_4	1.66	0.87	2.46E-02	2.92E-02
QPEA	1.14	0.36	1.64E-02	2.11E-02
QFT	1.37	0.59	1.76E-02	2.22E-02
mod5mils_65	1.04	0.26	1.54E-02	2.00E-02
Avg.	1.23	0.34	1.67E-02	2.13E-02

for each quantum computer. Collectively, all the benchmark circuits in Table II have been used to evaluate our proposed methods. Similar to the training dataset, we use PST and HF fidelity metrics for single- and multi-output quantum circuits, respectively.

We compare the generated quantum circuits using our RL model with Qiskit [1], t|ket [6], and noise adaptive [4] heuristic approaches and other RL models [11], [12] for quantum circuit mapping. t|ket uses a combination of BRIDGE (four CNOT gates) and SWAP gates, which can further reduce the gate count. To ensure a fair comparison, we 1) train all the RL models using the same dataset of random circuits and an equal number of episodes and 2) test all the RL models with benchmark circuits, each having the same initial qubit assignment for all RL models.

B. Experimental Results

1) *GNN-Based Fidelity Predictor*: Tables III and IV provide the inference time and the accuracy of our SGNN and MGNN fidelity predictors in the form of root-mean-square error (RMSE) for IBM Q Jakarta and IBM Q Guadalupe, respectively, for different quantum circuits. The output fidelity is computed by executing each quantum circuit on IBM Q Jakarta and IBM Q Guadalupe for 8192 shots. The tables

TABLE IV
INFERENCE TIME AND ACCURACY (RMSE) OF
SGNN AND MGNN FOR IBM Q GUADALUPE

Benchmark	Time (μ s)		RMSE	
	SGNN	MGNN	SGNN	MGNN
V_QAOA	1.98	1.16	1.55E-02	1.90E-02
tof_4	1.88	1.06	1.59E-02	1.94E-02
QPEA	1.11	0.29	8.43E-03	1.20E-02
QFT	1.75	0.93	1.16E-02	1.52E-02
mod5mils_65	1.01	0.19	8.22E-03	1.17E-02
mod5d1_63	1.15	0.33	7.78E-03	1.13E-02
mod_mult_55	2.90	2.08	1.03E-02	1.38E-02
MB	1.81	0.99	1.25E-02	1.60E-02
VQE	1.39	0.56	1.18E-02	1.53E-02
QAOA_1	6.35	5.53	2.00E-02	2.35E-02
QAOA_2	4.98	4.16	2.70E-02	3.05E-02
Avg.	2.39	1.57	1.36E-02	1.71E-02

show the high prediction accuracy of our SGNN and MGNN models for both quantum hardware. They also show that while the accuracy of our SGNN fidelity predictor is higher than the MGNN model, there is a significant reduction in the inference time of the MGNN compared to the SGNN that can reach up to 94% and 81% for IBM Q Jakarta and IBM Q Guadalupe, respectively. Although the LGNN models have fewer hidden layers than the SGNN model, the MGNN model aggregates information of more neighboring nodes from different blocks through the block dependency graph used in the GGNN, which provides good accuracy in a cost-effective way by reducing the number of exchanged messages between neighboring nodes. We also observe that the inference time of the MGNN model depends on the number of subcircuits in the block dependency graph and the size of each subcircuit, which explains the variation in the inference time reduction rate across different benchmarks of different circuit structures.

2) *Comparison With State-of-the-Art Noise-Aware Quantum Circuit Mapping*: We compare the properties of physical quantum circuits generated using our proposed RL (RL+SGNN and RL+MGNN), Qiskit [1], and t|ket [6] noise-aware quantum circuit mapping for IBM Q Jakarta and IBM Q Guadalupe quantum computers, including the output fidelity (F), the depth (D), and the number of CNOT gates (#CX) on average³ under the same qubit allocation in Tables V and VI, respectively. Our proposed RL framework always provides the highest output fidelity despite the circuit structure. Our RL+SGNN (RL+MGNN) approach improves the fidelity on average by a factor of $1.3 \times$ ($1.1 \times$) for IBM Q Jakarta, and $1.8 \times$ ($1.6 \times$) for IBM Q Guadalupe compared to the highest output fidelity provided by the other mapping approaches. We observe that the back-end error rates of IBM Q Guadalupe with a standard deviation in the range of 10^{-2} exhibit more variations in the error rate compared to IBM Q Jakarta with a standard deviation in the range of 10^{-3} . This implies that our proposed techniques are more effective for quantum hardware with high variation in error rates. Our results also

³We report the average output fidelity, the depth, and the number of CNOT gates of five different executions of quantum circuits in different calibration windows on IBM Q Jakarta and IBM Q Guadalupe.

TABLE V
AVERAGE COMPARISON OF QUANTUM CIRCUITS GENERATED USING DIFFERENT MAPPING TECHNIQUES AND EXECUTED ON IBM Q JAKARTA

Benchmrk	RL+SGNN			RL+MGNN			RL [11]			RL [12]			t ket> [6]			Qiskit [1]		
	F	D	#CX	F	D	#CX	F	D	#CX	F	D	#CX	F	D	#CX	F	D	#CX
mod5d1_63	76.66	29	24	74.22	35	28	54.86	41	24	55.53	44	37	66.00	25	18	49.67	41	41
4mod5-v0_20	60.45	35	25	55.76	31	28	46.14	38	39	43.80	39	34	49.66	18	12	28.42	46	47
4mod5-v0_19	58.38	71	35	56.92	59	39	52.52	59	48	46.71	62	50	53.60	37	28	46.42	64	56
V_QAOA	73.29	88	92	72.21	89	95	65.09	125	117	64.81	111	108	67.69	95	91	60.45	96	124
tof_4	58.81	99	54	51.63	99	57	37.72	100	77	32.75	117	108	45.80	86	57	32.53	104	91
QPEA	25.74	65	29	22.03	105	60	15.30	116	77	13.20	116	75	16.83	99	60	10.51	119	82
QFT	42.09	76	48	37.65	79	53	30.87	93	86	29.70	107	99	34.79	86	93	25.14	112	103
mod5mils_65	59.73	72	35	56.90	50	38	51.03	64	60	48.31	66	33	52.85	37	29	40.84	66	65
4mod5-v1_22	69.89	38	22	65.33	30	25	53.44	47	23	52.41	39	38	59.39	20	14	48.51	39	39
4gt13_94	32.00	121	63	29.13	114	66	17.38	116	119	17.46	108	85	24.86	72	51	15.97	123	127

TABLE VI
AVERAGE COMPARISON OF QUANTUM CIRCUITS GENERATED USING DIFFERENT MAPPING TECHNIQUES AND EXECUTED ON IBM Q GUADALUPE

Benchmrk	RL+SGNN			RL+MGNN			RL [11]			RL [12]			t ket> [6]			Qiskit [1]		
	F	D	#CX	F	D	#CX	F	D	#CX	F	D	#CX	F	D	#CX	F	D	#CX
V_QAOA	74.95	96	109	72.98	99	115	65.92	103	109	65.27	107	108	68.45	95	91	65.00	100	116
tof_4	56.69	100	61	49.59	105	64	35.16	94	57	37.27	102	49	44.65	96	54	28.30	114	69
QPEA	28.99	94	53	24.12	102	59	12.68	66	37	14.55	64	40	16.65	99	60	10.02	97	62
QFT	16.45	87	55	15.35	80	59	7.71	77	97	7.71	89	76	10.25	89	87	8.87	113	111
mod5mils_65	64.01	69	33	59.37	71	38	40.46	62	43	40.66	58	40	41.05	37	29	37.69	51	43
mod5d1_63	68.43	47	27	64.33	46	32	53.78	36	35	48.39	40	34	59.35	25	18	45.74	46	46
mod_mult_55	11.82	111	104	10.20	112	109	3.56	113	110	2.71	107	114	4.55	124	117	1.76	129	145
MB	6.38	161	119	4.97	155	126	1.35	144	128	1.28	163	141	1.84	100	100	0.69	169	149
VQE	73.06	52	51	67.82	59	56	47.25	64	57	46.63	89	62	51.21	93	69	32.82	143	88
QAOA_1	6.20	184	175	5.39	208	186	1.74	209	185	1.56	203	191	2.65	523	375	1.63	230	224
QAOA_2	18.79	106	88	15.28	104	92	9.27	82	74	10.37	109	83	11.19	303	210	9.51	111	101

TABLE VII
AVERAGE COMPARISON OF DIFFERENT QUANTUM CIRCUIT MAPPING TECHNIQUES FOR DIFFERENT BENCHMARKS EXECUTED ON IBM Q JAKARTA

Benchmark	RL+SGNN			RL+MGNN			NoiseAdapt [4]		
	F	D	#CX	F	D	#CX	F	D	#CX
mod5d1_63	70.53	34	26	64.69	47	31	56.21	47	39
4mod5-v0_20	59.59	28	22	56.02	30	24	48.70	35	28
4mod5-v0_19	54.88	59	37	52.78	52	42	48.83	78	40
V_QAOA	70.96	106	98	70.58	112	103	63.51	117	112
tof_4	48.09	89	61	44.54	92	64	38.38	112	70
QPEA	19.12	116	66	17.42	113	70	12.28	75	43
QFT	36.54	76	69	34.67	96	79	28.25	101	90
mod5mils_65	54.77	59	49	52.28	60	56	43.80	77	44
4mod5-v1_22	65.81	31	28	63.37	34	30	56.30	41	30
4gt13_94	28.19	98	76	26.07	99	79	19.48	130	73

show that minimizing the gate count or the circuit depth does not guarantee a significant improvement in the circuit output fidelity. It is worth mentioning that while our RL-based quantum circuit mapping approaches do not use any BRIDGE gate, they still outperform t|ket>.

Fig. 6(a) and (b) illustrates the best improvement in the output fidelity of quantum circuits generated using our proposed RL+SGNN and RL+MGNN compared to different other quantum circuit mapping approaches over a course of 5-day executions for IBM Q Jakarta and IBM Q Guadalupe, respectively. The figures show that our proposed RL+SGNN can increase the output fidelity by a factor of up to $2.5\times$ and $9.3\times$ for IBM Q Jakarta and IBM Q Guadalupe, respectively, while our proposed RL+MGNN can increase the output fidelity by a factor of up to $1.8\times$ and $6.3\times$ for IBM Q Jakarta and IBM Q Guadalupe, respectively, compared to the maximum output

fidelity achieved using the state-of-the-art quantum circuit mapping techniques. We report the growth factor in the output fidelity of RL+SGNN and RL+MGNN compared to the best output fidelity of other approaches for different benchmarks on the top of their bars in Fig. 6. We also observe as the number of CNOT gates and the circuit depth increase the effectiveness of our proposed models increases as well, i.e., a very high improvement in the output fidelity is observed for circuits with more than 100 CNOT gate count and circuit depth.

We expand our comparison to include Noise adaptive quantum circuit mapping (NoiseAdapt) [4] which enforces its own qubit allocation. Thus, we report on average the quantum circuit properties generated using our RL+SGNN, our RL+MGNN, and NoiseAdapt quantum circuit mapping for IBM Q Jakarta and IBM Q Guadalupe quantum computers

TABLE VIII
AVERAGE COMPARISON OF DIFFERENT QUANTUM CIRCUIT MAPPING TECHNIQUES FOR DIFFERENT BENCHMARKS EXECUTED ON IBM Q GUADALUPE

Benchmark	RL+SGNN			RL+MGNN			NoiseAdapt [4]		
	F	D	#CX	F	D	#CX	F	D	#CX
V_QAOA	71.95	110	109	68.36	98	110	62.96	101	124
tof_4	48.19	92	54	42.22	103	61	35.76	102	78
QPEA	20.00	65	33	17.11	79	39	11.91	108	64
QFT	14.55	77	62	12.87	81	62	5.75	97	80
mod5mils_65	53.33	64	46	49.68	69	50	40.34	55	46
mod5d1_63	58.95	39	34	56.20	41	38	48.24	39	33
mod_mult_55	8.21	118	112	6.59	113	115	3.51	130	130
MB	5.86	153	120	4.60	152	125	2.20	124	110
VQE	59.70	81	57	55.08	82	64	32.43	158	90
QAOA_1	4.20	197	195	3.86	208	204	1.58	325	283
QAOA_2	14.44	117	95	12.01	125	100	9.35	190	145

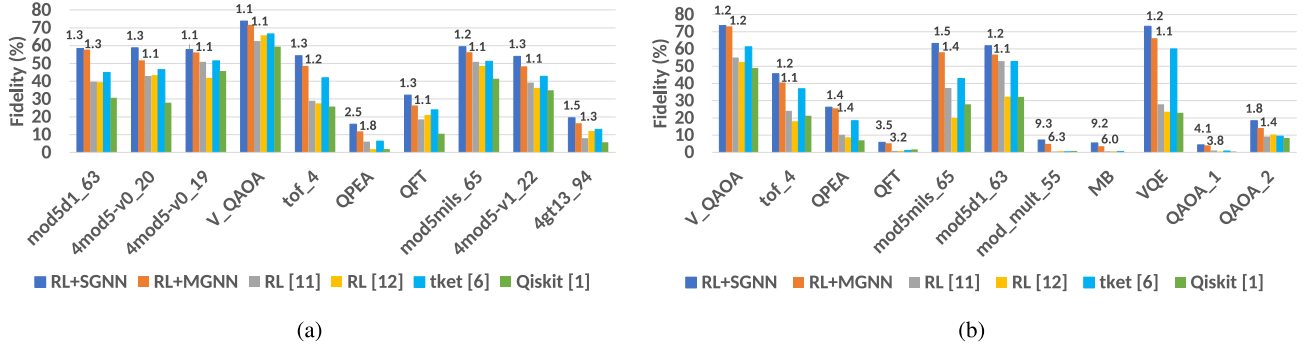


Fig. 6. Best fidelity comparison among different mapping techniques based on t|ket [6] initial qubit allocation for (a) IBM Q Jakarta and (b) IBM Q Guadalupe.

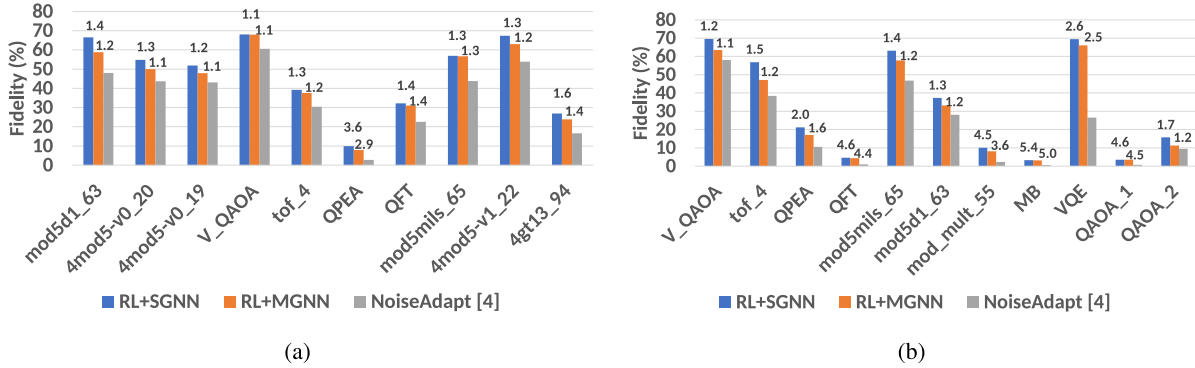


Fig. 7. Best fidelity comparison among different mapping techniques based on NoiseAdapt [4] initial qubit allocation for (a) IBM Q Jakarta and (b) IBM Q Guadalupe.

separately in Tables VII and VIII, respectively. Our RL techniques use the same qubit allocation as NoiseAdapt. We observe that our proposed RL+SGNN and RL+MGNN show even better improvement in the output fidelity compared to NoiseAdapt resulting in an increase in the output fidelity by a factor of (on average) $1.4\times$ ($1.3\times$) for IBM Q Jakarta, and on average by $1.8\times$ ($1.6\times$) for IBM Q Guadalupe. Fig. 7(a) and (b), which demonstrate the best results of our approaches also shows the superiority of our RL approaches compared to NoiseAdapt with the growth factor of the output fidelity for each quantum circuit benchmark.

3) *Comparison With Other RL-Based Quantum Circuit Mapping:* We compare the effectiveness of our RL techniques

in terms of the output fidelity of the quantum circuits in Tables V and VI for IBM Q Jakarta and IBM Q Guadalupe, respectively, and the runtime of our RL techniques in the testing phase (despite applying SGNN or MGNN during the training phase) with other recently proposed RL techniques that generate physical quantum circuits with the objective of either minimizing the gate count or the circuit depth [4], [6] in Table IX.

Tables V and VI show that while other RL techniques fail to provide better output fidelity under a fixed qubit assignment compared to heuristic-based approaches, such as mod5d1_63 and 4mod5-v0_19 executed on IBM Q Jakarta and tof_4 executed on IBM Q Guadalupe, our RL techniques based on

TABLE IX
CPU EXECUTION TIME (S) OF DIFFERENT MAPPING TECHNIQUES

Benchmark	mod5d1_63	4mod5-v0_20	4mod5-v0_19	V_QAOA	tof_4	QPEA	QFT	mod5mols_65	4mod5-v1_22	4gt13_92	mod_mult_55	MB	VQE	QAOA_1	QAOA_2
RL+GNN	0.3	0.3-0.4	0.4	0.8	0.3	0.4	0.3-0.4	0.2-0.3	0.3	0.3	67.2	0.8	1.8	87.2	80.1
[11]	0.4-0.5	0.4-0.5	0.5	0.8-0.9	0.5	0.4-0.5	0.6	0.5-0.6	0.4-0.5	0.4	77.3	1.4	4.5	91.4	83.3
[12]	0.2	0.3	0.3	0.7-0.8	0.3	0.4-0.5	0.4	0.3	0.3	0.7	62.8	0.7	1.2	84.6	76.4

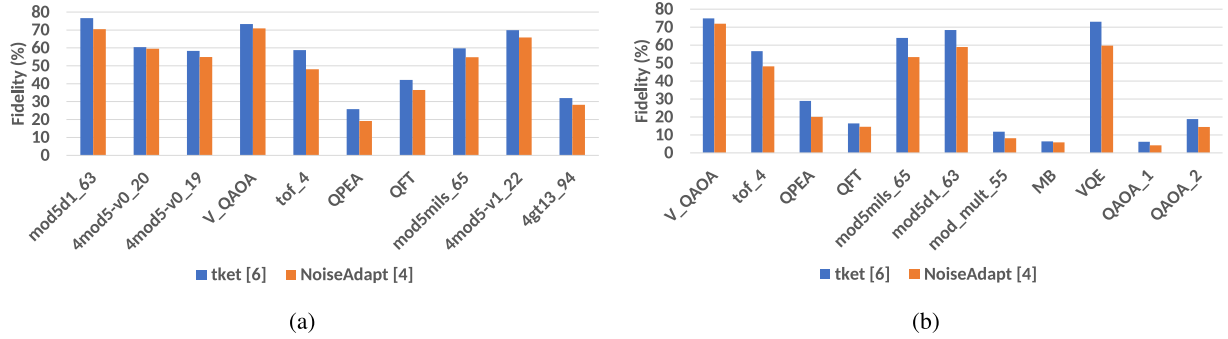


Fig. 8. Fidelity of RL+SGNN models using various initial qubit assignments. (a) IBM Q Jakarta. (b) IBM Q Guadalupe.

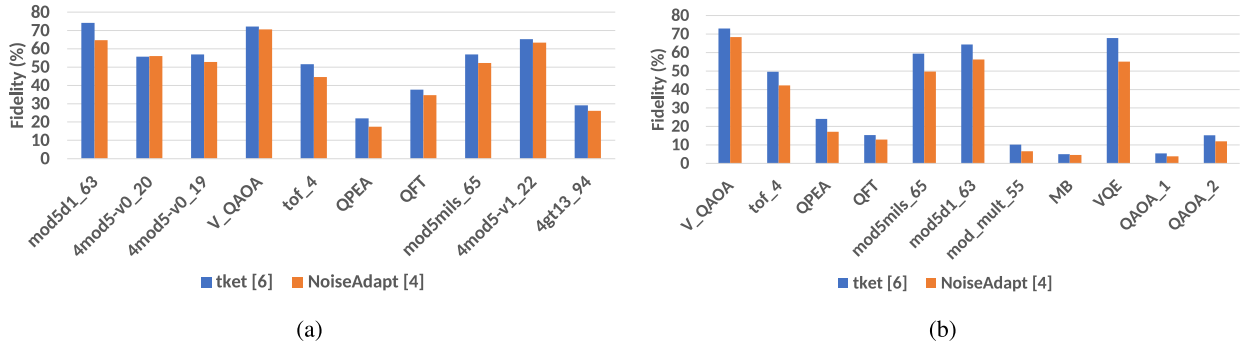


Fig. 9. Fidelity of RL+MGNN models using various initial qubit assignments. (a) IBM Q Jakarta. (b) IBM Q Guadalupe.

both SGNN and MGNN always provide the highest output fidelity.

In Table IX, we report the CPU execution time to generate physical quantum circuits for different benchmarks using our RL and other RL techniques listed above. Our proposed RL technique takes a relatively similar amount of time as the other RL techniques since the GNN fidelity predictor is used primarily during the training phase of our RL. The marginal increase in CPU execution time of our RL model compared to [12] is due to the features added for each gate in the current state, which are obtained from the back-end quantum hardware properties to avoid daily training.

4) *Effect of the Initial Qubit Allocation:* We analyze the sensitivity of our RL+SGNN and RL+MGNN techniques to different initial qubit allocations in Figs. 8 and 9, respectively. Specifically, Fig. 8(a) and (b) provides the average output state fidelity of the RL+SGNN approach for IBM Q Jakarta and IBM Q Guadalupe, respectively. Similarly, Fig. 9(a) and (b) shows the output state fidelity of the RL+MGNN approach for IBM Q Jakarta and IBM Q Guadalupe, respectively. Our

proposed RL methods consistently demonstrate higher fidelity for tket [6] initial qubit allocation than for NoiseAdapt [4]. The results indicate that the initial qubit allocation has a significant impact on the output fidelity of our RL models. Despite the sensitivity to the initial qubit allocation, our RL techniques consistently outperform existing mapping approaches in terms of the output fidelity under fixed qubit assignment. Thus, our proposed RL framework can benefit from the integration with efficient qubit allocation techniques (e.g., tket) to maximize the circuit output fidelity.

VII. DISCUSSION

We discuss some technical aspects of our proposed framework in the form of Q&A.

Question: How do you use the multi-GNN model for different quantum hardware with a fixed number of LGNNs?

Response: Each quantum architecture/computer has its own fixed number of LGNNs, which model the noise in predetermined qubit regions in the quantum hardware. To assess the

reliability of a physical quantum circuit, we first determine whether its qubits belong to any of the LGNN regions. If that is the case, we use the corresponding LGNN model to predict the circuit output fidelity. Otherwise, we partition the circuit into nonoverlapping subcircuits each consisting of qubits that belong to one qubit region.

Question: How do you use the GNN model to predict the fidelity of quantum circuits of different structures executed on different hardware?

Response: We build a GNN fidelity predictor per quantum hardware, which can be used under different noise profiles for different calibration windows. Thus, each quantum hardware has a different GNN model. We further leverage graph embedding, which enables fidelity prediction on unseen datasets of graphs of different numbers of nodes and edges, which represent quantum circuits with different structures.

Question: How does the trained RL framework learn to generate physical quantum circuit mapping across different calibration windows for unseen quantum circuits?

Response: The current state of the agent is encoded using graph embedding which captures both the circuit structure and the back-end properties of the quantum hardware during the calibration window in which the resulting physical quantum circuit will be executed. By interacting with the GNN predictor during the training phase, the agent learns the gap between the error rates of the quantum hardware and their impact on the circuit output fidelity.

Question: How do the proposed GNN-based models scale with the scale of quantum hardware size?

Response: As we scale the quantum hardware both training and inference raise scalability concerns. Training can benefit from utilizing application-dependent benchmarks (e.g., SupermarQ [37]) to provide diverse training datasets in a cost-effective way. Furthermore, the advancement at the hardware and the algorithmic levels of the GNN models can speed up not only the training but also the inference. Examples include using AI accelerators, applying hierarchical deep learning models, and combining heuristic optimizations with deep learning.

VIII. CONCLUSION

In this article, we propose a noise-adaptive quantum circuit mapping based on RL and GNN models to significantly improve the output fidelity of the quantum circuits executed on NISQ devices. Our RL agent learns a quantum circuit mapping policy to maximize the output fidelity by interacting with a GNN-based fidelity predictor that acts as quantum hardware and estimates the physical quantum circuit fidelity. We also propose a multilevel GNN-based fidelity predictor to enhance the scalability of our standard GNN model while maintaining good accuracy. We use benchmark circuits with diverse qubit numbers to evaluate our proposed framework. The experimental results show the effectiveness of our proposed RL framework for quantum circuit mapping compared to the state-of-the-art mapping techniques (i.e., Qiskit [1], t[ket] [6], and noise adaptive mapping [4]) and recently proposed RL-based

mapping approaches [11], [12] using real-world IBM quantum computers.

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Vedika Saravanan (Member, IEEE) received the B.S. degree in electrical engineering from the Sri Shakthi Institute of Engineering and Technology, Coimbatore, India, in 2016, the M.S. degree in electrical engineering from the Coimbatore Institute of Technology, Coimbatore, in 2018, and the Ph.D. degree in electrical engineering from The City College of New York, New York, NY, USA, in 2023.

She has published ten papers (journal/conference papers/book chapters) on quantum computing. Her research interests include quantum computing, hardware security, and design for testability. Her research is primarily focused on developing scalable quantum compilation approaches for reliable quantum computing.

Dr. Saravanan is the First-Place Winner of the CSAW'22 (AI versus Human) Hardware Security Competition in North America Region in 2022.



Samah M. Saeed (Member, IEEE) received the B.S. and M.S. degrees from the Department of Computer Science, Kuwait University, Kuwait City, Kuwait, in 2008 and 2010, respectively, and the Ph.D. degree from the Department of Computer Science and Engineering, New York University Tandon School of Engineering, Brooklyn, NY, USA, in 2015.

She is currently an Assistant Professor with The City College of New York, City University of New York, New York, NY, USA. Her research is funded by the Department of Energy and Google. Her research interests include quantum computing, hardware security, and testing integrated circuits.

Dr. Saeed received the Best Paper Award at IEEE VLSI Test Symposium 2011, the Pearl Brownstein Doctoral Research Award by NYU Polytechnic School of Engineering in 2013, and the TTTCs E. J. McCluskey Best Doctoral Thesis Award at the IEEE International Test Conference in 2014. She has served as a Session Chair/Moderator of conferences, including IEEE/ACM International Conference on Computer-Aided Design, IEEE European Test Symposium, IEEE International Conference on Computer Design, and International Symposium on Quality Electronic Design. She has also served as a PC Member on several program committees, such as ACM/IEEE Design Automation Conference, IEEE Computer Society Annual Symposium on VLSI, European Test Symposium, ACM/IEEE Great Lakes Symposium on VLSI, International Symposium on Quality Electronic Design, and IEEE International Symposium on Defect and Fault Tolerance in VLSI and Nanotechnology Systems.